Department of Civil Engineering

Engineering Science Course

Elements of Civil Engineering

CV113AT / CV123AT

UNIT II

Basic Materials of Construction

i) Bricks,
ii) Cement & mortars,
iii) Plain, Reinforced & Pre-stressed Concrete,
iv) Structural steel,
v) Construction Chemicals

Structural Elements of a building

i) foundation,
ii) plinth, lintel, chejja,
iii) masonry wall,
iv) column, beam, slab
v) staircase
vi) plinth area, carpet area, floor area ratio,
vii) local building byelaws.

BRICKS

Bricks - Introduction

Bricks are a type of building material typically made of concrete, sand, lime, or clay. They are generally used to construct walls, pavements, and other types of architecture. Bricks can be produced in a variety of shapes and types depending on the materials used to make them and the use for which they are intended. It is because of their sturdiness, strength, and fire resistance, they continue to be a popular building material.

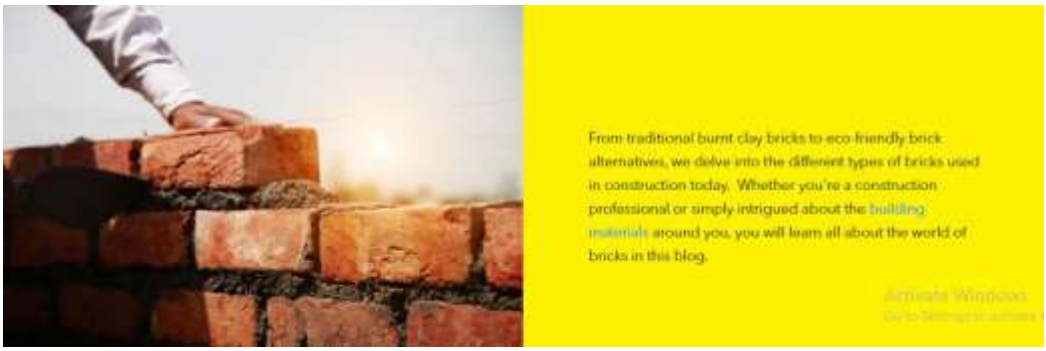
On the basis of quality, Bricks are of the following kinds:

1. First Class Brick: The size is standard. The color of these bricks is uniform yellow or red. It is well burnt, regular texture, uniform shape. The absorption capacity is less than 10%, crushing strength is, 280kg/cm^2 (mean) where it is 245 kg/cm^2 (minimum).

2. Second Class Brick: The size is standard, color is uniform yellow or red. It is well burnt, slightly over burnt is acceptable. It has a regular shape; efflorescence is not appreciable. The absorption capacity is more than 10% but less than 15%. Crushing strength is 175kg/cm^2 (mean) where the minimum is 154 kg/cm^2

3. Third Class Brick: The shape and size are not regular. The color is soft and light red colored. It is under burnt, slightly over burnt is acceptable. It has extensive efflorescence. The texture is non-uniform. The absorption capacity is more than 15% but less than 20%.

4. Fourth class bricks: These are over-burnt or under-burnt and have poor quality. They are unsuitable for any construction purposes as they may have cracks, distortions, and other defects.



Reference: https://www.ultratechcement.com/

Properties of brick:

i) The color of bricks should be bright and uniform.

ii) They should be well burned and having smooth surfaces and sharp edges.

iii) Thermal conductivity of bricks should be less and they should be sound proof.

iv) They shouldn't absorb more than 20% by weight when we placed it in water.

v) When we struck two bricks together, ringing sound should be delivered.

vi) Structure of bricks should be homogeneous and uniform.

vii) The bricks should not break when we dropped it form 1m height.

viii) There should not be any scratch left on the brick when we scratched with finger nail.

ix) There should not be any white deposits on brick, when we soaked it in water for 24 hrs.



First Class Brick



Second Class Brick



Third Class Brick



Fourth Class Brick

The standard brick size in India is 190 mm x 90 mm x 90 mm, as per the BIS guidelines. When the mortar is thick, the bricks' dimensions shift to 200 mm x 100 mm x 100 mm, which is sometimes referred to as the nominal size of the modular brick.

CEMENT

Cement – Introduction

Cement is defined as a binding agent that is used to bind various construction materials. Given its adhesive and cohesive properties, it is an essential ingredient of concrete and mortar. Cement is mixed with water to form a paste that binds aggregates like sand or crushed rocks. Calcium, silicon, iron and aluminium compounds are closely ground to form a fine powdered product – cement.

The usage of cement in various forms has been advent through the years. In the ancient times, crushed pottery, volcanic ashes, and other items were used as cement. In 1824, Joseph Aspdin created the precursor to modern-day cement – Portland cement

Hydraulic cement is formed by the reaction of powdered cement with water. You can use it for all types of construction, including underwater construction projects. Non-hydraulic cement sets and becomes adhesive due to carbonation. You can use it for various construction projects, except underwater construction projects.

Hydraulic cement is the most commonly used cement. Portland cement is a type of hydraulic cement that's been a preferred choice for architects, engineers, and constructors. The primary reason behind its popularity is its ability to harden quickly. Per the Bureau of Indian Standards (BIS), predominately we use OPC 53 grade cement for construction purpose:

Properties of Cement:

i) Initial setting time $\leq 30\ minutes$

ii) Final setting time $\not> 10\ hours$

iii) 3-day compressive strength $\not< 16\ N/mm^2$

iv) 7-day compressive strength $\not> 22\ N/mm^2$

v) 3-day tensile strength $\simeq 2\ N/mm^2$

vi) 7-day tensile strength $\simeq 2.5\ N/mm^2$

vii) When sieved on I.S. 90 micron sieve, residue $\not> 10\%$

viii) Expansion in Le-chatlier's test $\not> 19mm$

Types of Cement

Cement is mainly classified into two categories depending on the hardening and setting mechanism

i) Hydraulic Cement

ii) Non-hydraulic Cement

Along with these main types, depending on the composition and characteristics there are:

1. Ordinary Portland Cement (OPC)
2. Portland Pozzolana Cement (PPC)
3. Rapid Hardening Cement
4. Quick Setting Cement
5. Low Heat Cement
6. Sulphate Resisting Cement
7. Blast Furnace Cement
8. High Alumina Cement
9. White Cement
10. Colored Cement



Typical Cement Bag

Ordinary Portland Cement (OPC) When Portland cement clinker is mixed with gypsum, it forms OPC. OPC 53 with a compressive strength of 53MPa at 28 days is termed as OPC 53. OPC 53 is mostly used for precast concrete, prestressed concrete, long span structures like bridges, tall buildings. The most widely used type, suitable for general construction. It sets and hardens with water and is known for its strength and durability.

Portland Pozzolana Cement (PPC) is a type of blended cement that incorporates pozzolanic materials—natural such as volcanic ash or fly ash or artificial substances. This combination enhances various properties of the cement, making it suitable for a wide range of construction applications, pozzolanas react with calcium hydroxide in the presence of water, enhancing resistance to chemical attacks, reducing heat of hydration, improved durability and low permeability.

Rapid Hardening Cement is a type of cement specifically formulated to set and gain strength much faster than ordinary Portland cement. It is known to reduce construction time with accelerated strength development at almost similar workability as OPC. Exclusively preferred for pre-cast construction, emergency repairs, thin concrete sections and cold weather concreting

Quick Setting Cement is a specialized type of cement designed to set rapidly, typically within a few minutes to a couple of hours after mixing with water, thereby resulting in fast setting time with reduced water requirement. It is known for early strength development without reducing workability, but results in a slightly higher heat generation. Used typically for repairs, anchorage and construction of temporary structures.

Low Heat Cement is a type of cement designed to produce less heat during the hydration process compared to ordinary Portland cement. This characteristic makes it particularly useful in specific construction scenarios, thereby resulting in reduced heat generation and slow setting time. Used typically in Mass concrete, cold weather construction and constructions of roads/pavements.

Sulphate Resisting Cement (SRC) is specifically designed to withstand the damaging effects of sulfate attacks, which can lead to structural deterioration in certain environments. It contains low C3A and hence has enhanced durability, also known to possess low permeability with high strength which ensures long term durability in severe exposure conditions. Suited specifically for foundations / retaining walls, marine / coastal structures and chemical related industry.

Blast Furnace Cement (BFC) is a type of blended cement made by mixing granulated blast furnace slag (a byproduct of iron production) with ordinary Portland cement. The slag enhances long term performance by reducing permeability and thus improves durability. Also, with low heat of hydration the cement is known to be eco-friendly, cost-effective and sustainable.

High Alumina Cement (HAC) is a type of cement that is rich in alumina, typically containing around 30-50% alumina (Al_2O_3). It is known for its rapid strength development and high resistance to high temperatures and chemical attacks. Cement is known for early strength, high temperature resistance, thermal stability and fast construction. Specifically suited for Refractory applications, marine structures and repair works.

White Cement is a type of cement that is characterized by its white color, achieved by using raw materials with low iron content. It is primarily used for aesthetic and architectural purposes. The raw materials are typically high-purity limestone, kaolin, and other materials that minimize iron and manganese oxides. The primary distinguishing feature is its bright white color, which provides a clean and attractive finish. It is highly reflective and has compatibility with pigments. Specifically used for architectural / pre-cast concrete and interior finishes.

Colored Cement is a type of cement that has been specifically formulated to achieve various colors, typically through the addition of pigments. This allows for enhanced aesthetic appeal in concrete and construction applications. Pigmentation is achieved by adding inorganic or organic pigments to ordinary Portland cement, enabling a range of colors from subtle hues to vibrant shades. It is compatible with standard cementitious materials and can be mixed with aggregates to create colored concrete with decoration as primary application.

MORTAR

Mortar – Introduction

Mortar is a homogenous mixture of cement, sand and water. Different types of mortars are used in masonry construction based on their applications, binding materials, strength, bulk density and their purposes.

According to 'Frederick S. Merritt', (Author of Building Design and Construction Handbook), mortars are composed of a cementitious material, fine aggregate, sand, and specific amount of water. Mortar can be used for a number of purposes such as plastering over bricks or other forms of masonry, for flooring etc., and with the addition of coarse aggregate, it can also be used to make concrete. Cement mortar also provides a superior medium to create a smooth surface on walls made from bricks or other forms of masonry.

Properties of Mortar:

i) Adhesion: Mortar must bond well with the masonry units and the substrate to ensure structural integrity.

ii) Workability: Should be easy to mix, apply, and shape, allowing for smooth application during construction.

iii) Durability: Must withstand environmental conditions, including moisture, temperature variations, and chemical exposure.

iv) Flexibility: Some types of mortar allow for slight movement without cracking, accommodating structural shifts.

v) Strength: The compressive and tensile strength of mortar is important for the load-bearing capacity of masonry structures.

vi) Water Retention: Adequate water retention is crucial for proper curing and strength development.

Types of Mortar:

i) Cement Mortar - Composition: Cement, sand, and water.

Properties: Strong and durable; suitable for general construction and load-bearing applications.

Uses: Bricklaying, blockwork, and stone masonry.

ii) Lime Mortar - Composition: Lime, sand, and water.

Properties: More flexible and breathable than cement mortar; allows for movement without cracking.

Uses: Traditional masonry work, restoration projects, and applications where flexibility is required.

iii) Gypsum Mortar - Composition: Gypsum plaster and water.

Properties: Quick-setting; not suitable for outdoor use due to water sensitivity.

Uses: Interior applications, such as plastering walls and ceilings.

iv) Surkhi Mortar - Composition: A mixture of lime, sand, and finely crushed brick (surkhi).

Properties: Offers good workability and is often used in traditional construction.

Uses: Brick masonry, especially in regions where traditional materials are preferred.

v) Polymer-Modified Mortar - Composition: Cement, sand, and polymers.

Properties: Enhanced adhesion, flexibility, and water resistance.

Uses: High-performance applications, including tile setting and repairs.

vi) Refractory Mortar - Composition: High-alumina cement, fireclay, and sand.

Properties: Resists high temperatures; suitable for furnaces and chimneys.

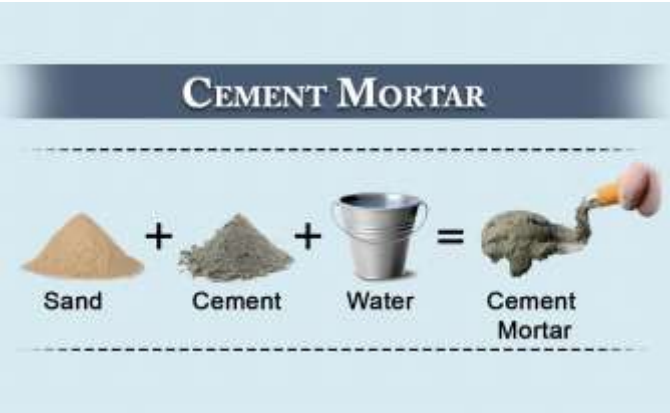
Uses: Lining for kilns, fireplaces, and other high-temperature applications.

vii) Masonry Cement - Composition: A blend of Portland cement, plasticizers, and other additives.

Properties: Improved workability and water retention.

Uses: Brick and block laying, offering a balance between strength and workability.

Each type of mortar has its unique properties and is suited for specific applications, ensuring the right choice can enhance the durability and performance of masonry structures.



Ingredients of conventional cement mortar

Proportions of Mortar: commonly recommended for different works

Masonry Construction:

i) For ordinary masonry work with brick/ stone as a structural unit. – **1:3** to **1:6**

ii) For reinforced brick work – **1:2** to **1:3**.

iii) For all work in moist situations – **1:3**

iv) For Architectural work – **1:6**

v) For Load Bearing structures – **1:3 or 1:4**

Plaster Work:

i) For External Plaster and Ceiling Plaster – **1:4**

ii) Internal Plaster (If sand is not fine i.e. Fineness Modulus> 3) – **1:5**

iii) For Internal Plaster (if fine sand is available) – **1:6**

Curing of Cement Mortar Cement gains strength with hydration. So, it is necessary to see that the mortar remains wet until hydration occurs. After placing the mortar/concrete, the process of ensuring sufficient moisture for hydration is called curing. Curing is ensured by spraying water. Generally, curing begins 6–24 hours after using mortar. Initially, more water is required for hydration, which can be reduced gradually. Curing for cement mortar is recommended for 7 days.

CONCRETE

Concrete – Introduction

Concrete is a composite material composed of aggregate bonded together with a fluid cement that cures to a solid over time. Concrete is the second-most-used substance in the world after water, and is the most widely used building material. The different forms of concrete include: Plain Concrete, Reinforced Concrete, and Pre-stressed Concrete

Plain Cement Concrete (P.C.C.) is the mixture of cement, fine aggregate (sand) and coarse aggregate without steel. PCC is an important component of a building which is laid on the soil surface to avoid direct contact of reinforcement of concrete with soil and water.

Material Used in Plain Cement Concrete

i) Coarse Aggregate Coarse aggregate used in the PCC must be of hard broken stone of granite or similar stone, free from dust, dirt and other foreign matter. The stone shall be 20 mm in size and smaller. All the coarse material should be retained in a 5mm square mesh and should be well graded so that the voids do not exceed 42%.

ii) Fine Aggregate Fine aggregate shall be of coarse sand consisting of hard, sharp and angular grains and shall pass through a screen of 5 mm square mesh. Sand shall be of standard specifications, clean and free from dust, dirt and organic matter.

iii) Cement Portland Pozzolana cement (P.P.C) is normally used for plain cement concrete. It should conform to the specifications and shall have the required tensile and compressive stresses and fineness.

iv) Water used shall be clean and reasonably free from injurious quantities of deleterious materials such as oils, acids, alkalis, salts and vegetable growth. Generally, potable water shall be used having a pH value not less than 6.

Proportioning of Plain Cement Concrete 1. The proportioning is done based on the requirement or given specification. Generally 1:2:4 or 1:3:6 mix is used. Reinforced concrete: incorporates steel reinforcement bars (rebar) or steel fibers into the concrete mix to enhance its tensile strength.

Reinforced Cement Concrete (R.C.C) is the combination of ordinary concrete with the steel reinforcement to increase its compressive and tensile strength to a great extent.

Nature of Reinforced Cement Concrete: The main principle in the preparation of the reinforced cement concrete is to make a structural material in which:

i) Steel serves the purpose of bearing the main tensile stresses;

ii) concrete bears the main compressive forces, both acting in complete unison;

Some common types of reinforcement are:

i) Mild Steel Bars: This steel bar used as reinforcement can be commonly bent easily without cracking at the bends.

ii) Hot Rolled Bars and Cold Worked Bars: Hot Rolled Bars has a characteristic strength in tension which is almost double than that of mild steel bars.
They can be bent by heating (up to 100°C) without developing any defects.
Similarly, the cold worked steel bars come in twisted or stretched forms having elongated ribs or such structures along their length.

iii) Steel Fabric: This is made from a variety of bars and wires. These may include plain round wires, indented and deformed wires, deformed steel bars of cold-worked type.

Pre-stressed concrete is a form of reinforced concrete in which internal stresses are introduced before the concrete is subjected to external loads, enhancing its performance.

Properties:

i) Increased Load Capacity: Can support greater loads and spans compared to conventional reinforced concrete.

ii) Reduced Cracking: The pre-stressing helps to counteract tensile forces, minimizing the risk of cracking.

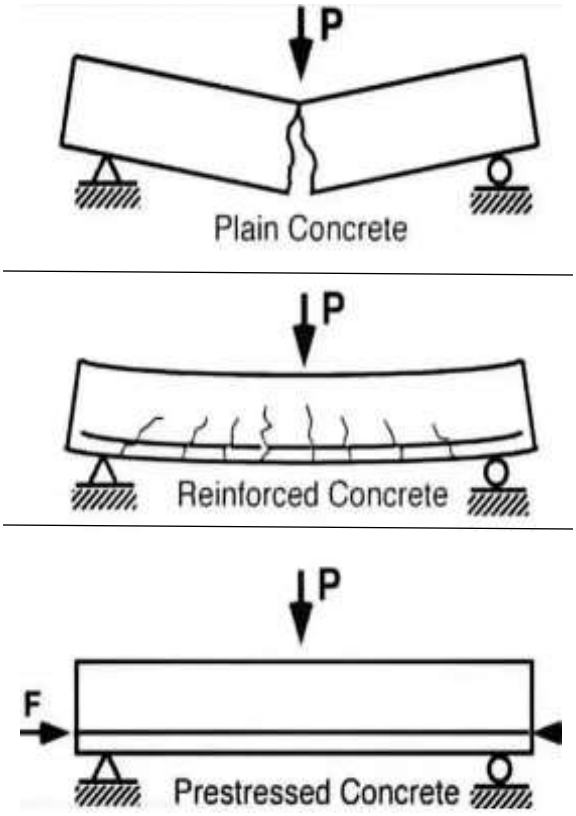
iii) Efficient Use of Materials: Allows for thinner sections and longer spans, optimizing material usage.

Applications:

i) Long-Span Structures: Used in bridges, parking garages, and large commercial buildings.

ii) Tanks and Silos: Effective in applications requiring high resistance to pressure.

iii) Highway and Railway Bridges: Common in infrastructure projects due to its ability to span long distances without support.



Reference: https://x.com/CivilEngCo



Reference: Google Image

Structural Steel – Introduction

Structural Steel is a special kind of Steel. It is used for construction purposes. Due to its rigidity and high strength-to-weight ratio, structural Steel is mainly employed in buildings. Structural Steel is used in houses, warehouses, airplane hangars, educational facilities, bridges, stadiums, etc.

Structural Steel is Steel that contains carbon, not more than 2.1%. These are also called Carbon Steel, and structural Steel typically has a carbon content of less than 0.6%.

Properties of Structural Steel:

i) Density: The density of Structural Steel is 7750 to 8100 kg/m3.

ii) Young's Modulus of Elasticity: Typical values for structural steel range from 190-210 GPa

iii) Poisson's ratio: For structural Steel, the acceptable value ranges from 0.27 to 0.3.

iv) Tensile strength: Structural Steel has high tensile strength, so it is preferred over other construction materials.

v) Yield strength: The yield strength, also known as the yield point, is the stress at which an object permanently deforms. When stress is removed, it does not revert to its former shape. Carbon structural steel has a yield strength ranging from 187 to 758 MPa. The values of structural Steel constructed of alloys range from 366 to 1793 MPa.

vi) Shear strength: The shear strength of steel structure is specified at the failure under shear stress, and it is about 0.57 times the yield stress of structural Steel.

vii) Hardness: The resistance of an object to shape change when force is applied is referred to as hardness. There are three different types of hardness tests. Scratch, indentation, and rebound are all terms used to describe the process of scratching and indenting, and the hardness of structural Steel manufactured with alloys ranges from 149 to 627 kg. Carbon structural steels have a weight range of 86 to 388 kg.

Types of Structural Steel:

i) Carbon steel: Steel in which the carbon content is upto 2% is known as carbon steel. The Specified ultimate tensile strength is 410 to 440 MPa, and the yield strength is 350 to 400 MPa

ii) High-strength carbon steel: These steels are used in structures such as transmission lines and microwave towers. The specified ultimate tensile strength is 480 to 550 MPa, and the yield strength is 350 to 400 MPa.

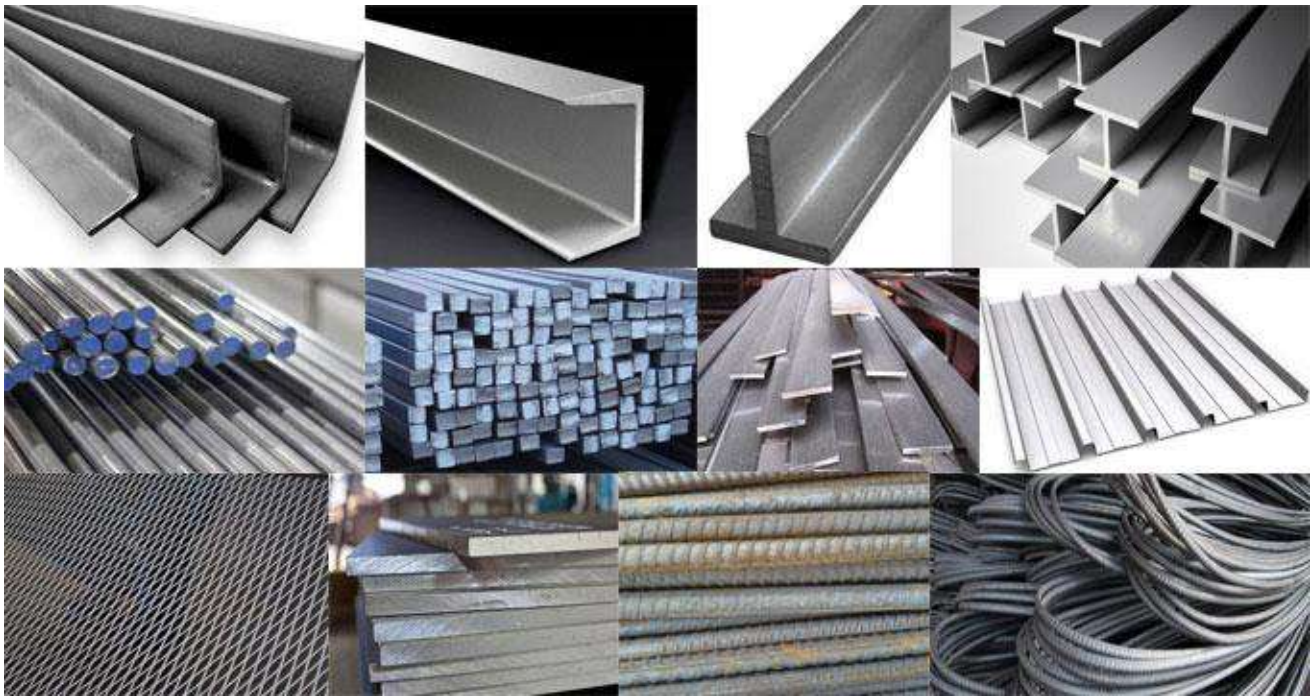
iii) Weathering Steel: These are corrosion-resistant Steel and are often not Painted. The specified ultimate tensile strength is 480 MPa, and the yield strength is 350 MPa.

iv) Fire-resistant Steel: These steels are also known as thermo mechanically treated (TMT) steel and are used where the structures are more prone to fire.

Types of Steel Sections

Structural steel members are fabricated in factories according to their intended use. Continuous casting molds are used to cast rolled steel parts with no joints. The following sections describe the various shapes and forms of rolled steel sections.

i) Rolled Steel I-sections (Beam sections).
ii) Rolled Steel Channel Sections
iii) Rolled Steel Tee Sections
iv) Rolled Steel Angles Sections
v) Rolled Steel Bars
vi) Rolled Steel Tubes
vii) Rolled Steel Flats
viii) Rolled Steel Sheets



Representation of common steel sections

Construction chemicals - Introduction

Construction chemicals have always been playing important roles in virtually all sorts of construction projects, be it industrial projects, residential building projects, commercial building projects and so on. These chemicals are often used in various elements of projects in order to achieve various important qualities such as workability, durability etc. Construction chemicals exist in many varieties from a large number of manufacturers worldwide

Construction methods and techniques have evolved greatly over the past centuries, all over the world. Previously, the materials and methods for construction were basic, such as simple brick/stone and mortar of some kind. However, as construction technology evolved, the trend of construction additives became more prominent and prevailing, leading to numerous types of construction chemicals.

i) Concrete curing compounds: These compound consists essentially of waxes, natural and synthetic resins, and solvents of high volatility at atmospheric temperatures. The compound forms a moisture retentive film shortly after being applied on a fresh concrete surface.

ii) Polymer Bonding Agent is an aqueous emulsion of a polymer and chemical admixtures. It is designed for use as a bonding agent with concrete and cement-based products in interior or exterior applications.

iii) Mould release agents come in handy when you have materials that are shaped and constructed in moulds. Without the releasing agent, your mould may become damaged or even break when it is time to remove it.







iv) Form release agents: These compounds are applied on the inner surfaces of forms, not only facilitate stripping of formwork but also render concrete surfaces smoother. They also help enhance the life-span of the forms. Form releasing agents can be oil based, resin based, water based, organic chemical based

v) Concrete floor hardeners These are chemicals added in floor concrete in order to render it denser and more durable. They also usually enhance chemical resistance, impact & abrasion resistance, waterproofing capability etc. besides reducing dusting







vi) Tile fixing: Tile fixers and tile adhesives form the backbone of your home. A quality tile fixer connects all your tiles together, to create a beautiful canvas from individual pieces. Tile fixing products are used for floorings, bathtubs, washbasins, kitchen tops and any other area where two surfaces need to stick together.







vii) Waterproofing chemicals

These chemicals can be quite useful when a structure's waterproofing capability is to be given a boost which is especially required for structures constantly dealing with liquids. There are many varieties.

viii) Adhesives

These construction chemicals are readily used in all sorts of projects, be it commercial, residential, industrial etc. construction projects. Adhesives are expected to have strong bonding capacity besides good waterproofing, weatherproofing etc. qualities.







BUILDING STRUCTURAL ELEMENTS

FOUNDATIONS

Foundation - Introduction

A foundation is an engineering term for the part of a structure that connects it to the soil and transmits loads from the building to the earth first. In general, foundations are categorized as shallow or deep. Geotechnical engineering principles such as soil mechanics & rock mechanics are used in foundation engineering to create the foundation components of buildings.

Functions of foundation:

i) Spread out the structure's weight across a vast region to prevent overwhelming the ground's soil without exceeding the bearing capacity of soil.

ii) To anchor the structure in the event of various natural calamities, including earthquakes, floods, droughts, frost heaves, storms, and wind.

iii) A flat surface to facilitate building. To securely anchor the building, improving stability and preventing overloading.

iv) To stop the supporting structure from swaying to the side.

Types of Foundation:

i) Shallow Foundation

According to Terzaghi, a shallow foundation is one whose width is greater than its depth. (i.e. $D_f/B \leq 1$). Such a foundation transmits the load laterally to the upper strata of the earth & is generally provided to the lightweight structures. It is preferred when foundation soil has sufficient bearing capacity at shallow depth. When the sum of areas covered by each isolated footings is more than 50% of the total area of the foundation, mat foundation is adopted.

Classification of Shallow Foundation:

i) Wall Footing

ii) Isolated column or Column Footing

iii) Combined Footing

iv) Cantilever (Strap) Footing

v) Mat (Raft) Foundation

ii) Deep Foundation:

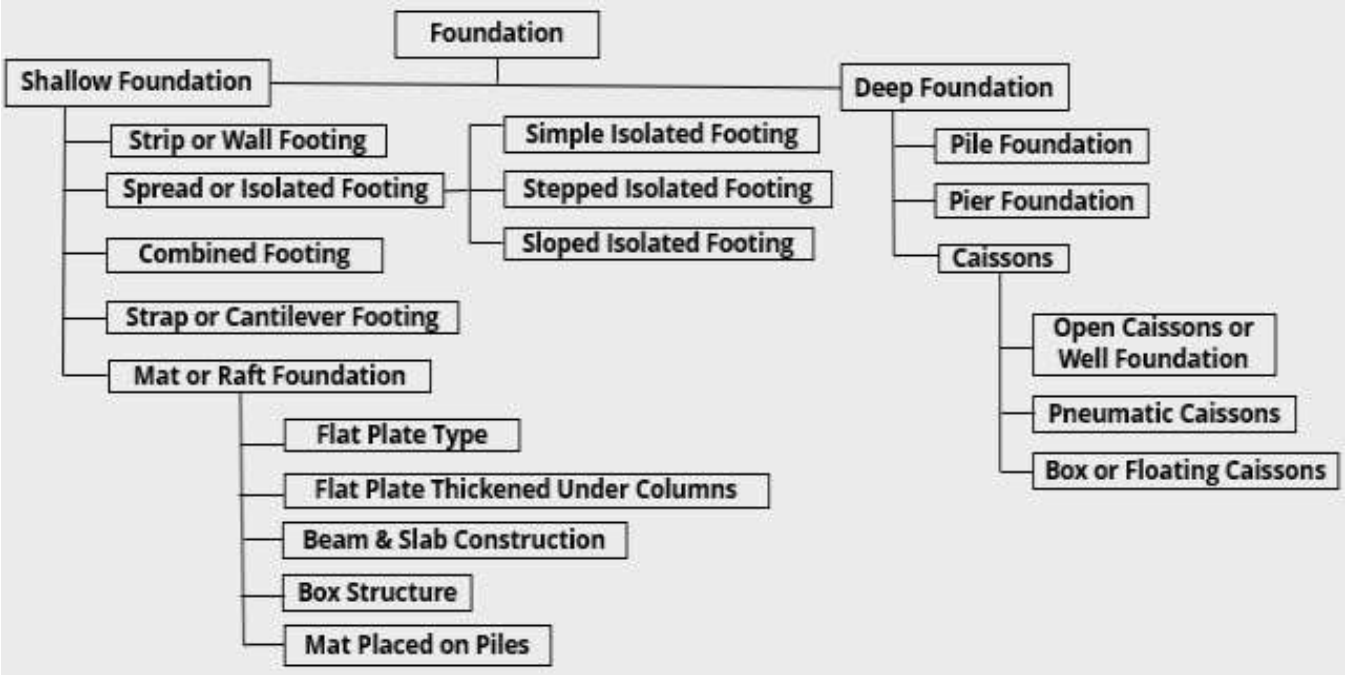
Deep Foundation are those foundations in which the depth of the foundation is greater than its width ($D_f > B$). The D_f/B ratio is usually $4-5$ for deep foundation. Unlike shallow foundation, the deep foundation transmits the load of the superstructure vertically to the rock strata lying deep. Deep foundations are used when the shallow foundation cannot support the load of the structure.

Classification of Deep Foundation:

i) Pile Foundation

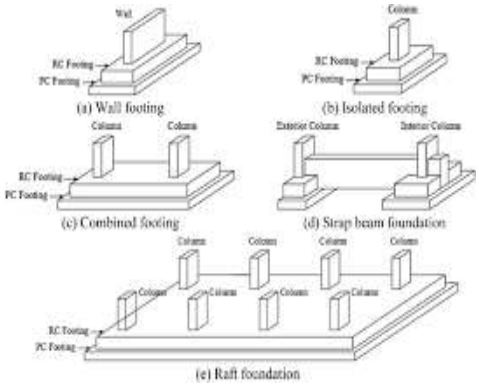
ii) Pier Foundation

iii) Caisson / Well Foundation

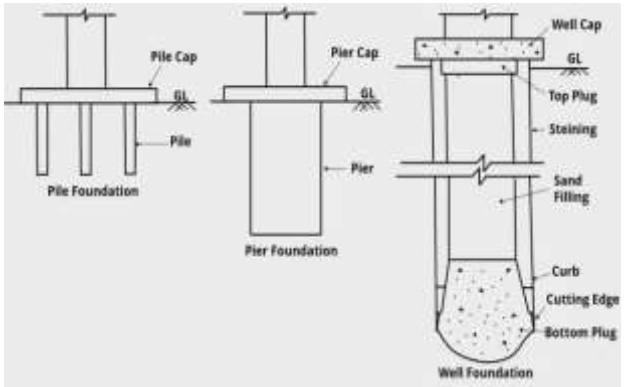


Classification of Foundation

Reference: https://www.dhakalvivek.com.np/



Types of shallow foundation



Types of deep foundation

Reference: Google Image

PLINTH, LINTEL & CHEJJA

Plinth level - Introduction

The level at which Substructure ends and superstructure starts is called Plinth level. It is the part of the superstructure between natural ground level and Finished floor level. the plinth is provided to restrict the seepage of stormwater and rainwater into the building.

The plinth height is in between 300mm – 450 mm from ground level. It is recommended that the minimum plinth height of 150 mm is adopted from the top of the road. The plinth is the base or platform on which a building or structure stands. It is usually a raised section that provides stability and elevation.

Damp proof course (DPC) is laid on Plinth level. The purpose of applying DPC is to restrict the movement of moisture through walls and floors. In Simple when you climb 3-4 steps to reach the building ground level is called Plinth height.

Functions of Plinth:

i) Support: Distributes the load of the structure evenly to the foundation.

ii) Protection: Raises the structure above ground level to protect against moisture and pests.

iii) Aesthetic Appeal: Can enhance the visual aspect of a building's design.

Significance of Plinth:

i) Ensures the longevity and stability of the structure by providing a solid base.

ii) Helps prevent water ingress and damage to the lower parts of the building

Sill level - Introduction

Sill level or Window Sill level: The level between the base portion of the window and portion of the floor above ground level (upwards) is called Sill level. Mortar bed or concrete bed is laid at the base of the window. The height of sill level depends upon the type of room for bedroom & bathroom the height may kept around minimum 1100mm due to privacy concrens and in the living room the window sill level is kept at minimum 600-650mm from the floor level. It is recommended that the minimum sill level height of 44 inches.

Lintel level - Introduction

A lintel is a beam placed across the openings like doors, windows etc. in buildings to support the load from the structure above. Lintel is provided above the door and window to transfer the upward wall load to the surrounding wall. Lintel is generally made up of Reinforced concrete or cement mortar. The width of lintel beam is equal to the width of wall, and the ends of it is built into the wall. Lintels are classified based on their material of construction.

Functions of lintel:

i) Load Distribution: Transfers the load from the structure above the opening to the walls on either side.

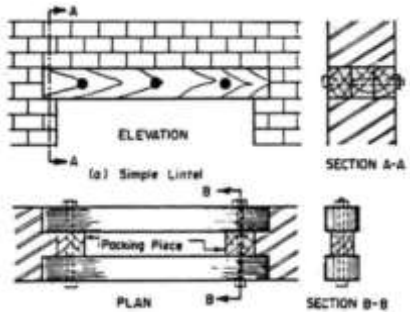
ii) Structural Integrity: Maintains the stability and integrity of the wall by preventing sagging or collapse.

Types of Lintel used in Building Construction

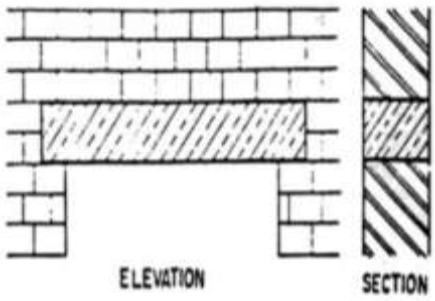
i) Timber Lintel In olden days of construction, Timber lintels were mostly used. But now a days they are replaced by several modern techniques, however in hilly areas these are using. The main disadvantages with timber are more cost and less durable and vulnerable to fire. If the length of opening is more, then it is provided by joining multiple number of wooden pieces with the help of steel bolts which was shown in fig (a). In case of wider walls, it is composed of two wooden pieces kept at a distance with the help of packing pieces made of wood. Sometimes, these are strengthened by the provision of mild steel plates at their top and bottom, called as flitched lintels

ii) Stone Lintel These are the most common type, especially where stone is abundantly available. The thickness of these are most important factor of its design. These are also provided over the openings in brick walls. Stone lintel is provided in the form of either one single piece or more than one piece. The depth of this type is kept equal to 10 cm / meter of span, with a minimum value of 15 cm. They are used up to spans of 2 meters. In the structure is subjected to vibratory loads, cracks are formed in the stone lintel because of its weak tensile nature. Hence caution is needed.

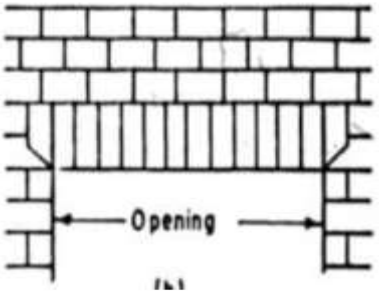
iii) Brick Lintel These are used when the opening is less than 1m and lesser loads are acting. Its depth varies from 10 cm to 20 cm, depending up on the span.



Timber lintel



Stone lintel

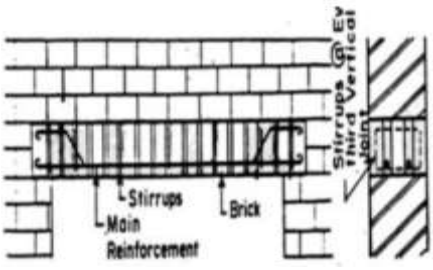


Brick lintel

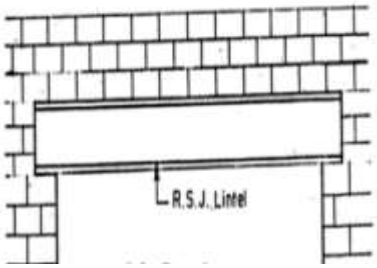
iv) Reinforced Brick Lintel These are used when loads are heavy and span is greater than 1m. The depth of reinforced brick lintel should be equal to 10 cm or 15 cm or multiple of 10 cm. the bricks are so arranged that 2 to 3 cm wide space is left length wise between adjacent bricks for the insertion of mild steel bars as reinforcement. 1:3 cement mortar is used to fill up the gaps. Vertical stirrups of 6 mm diameter are provided in every 3rd vertical joint. Main reinforcement is provided at the bottom consists 8 to 10 mm diameter bars, which are cranked up at the ends

v) Steel Lintel These are used when the superimposed loads are heavy and openings are large. These consist of channel sections or rolled steel joists. We can use one single section or in combinations depending up on the requirement.

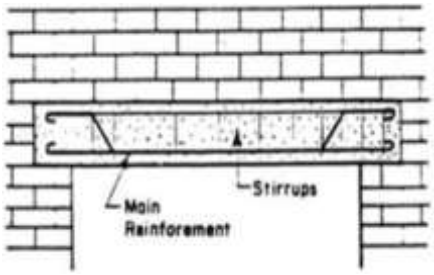
vi) Reinforced Cement Concrete Lintel at present, the lintel made of reinforced concrete are widely used to span the openings for doors, windows, etc. in a structure because of their strength, rigidity, fire resistance, economy and ease in construction. These are suitable for all the loads and for any span. The width is equal to width of wall and depth depends on length of span and magnitude of loading.



Reinforced brick lintel



Steel lintel



Reinforced concrete lintel

Chejja level - Introduction

A chejja, also known as a chajja or eave, is a projecting roof overhang that extends beyond the walls of a building. It serves both functional and aesthetic purposes.

Functions of chejja:

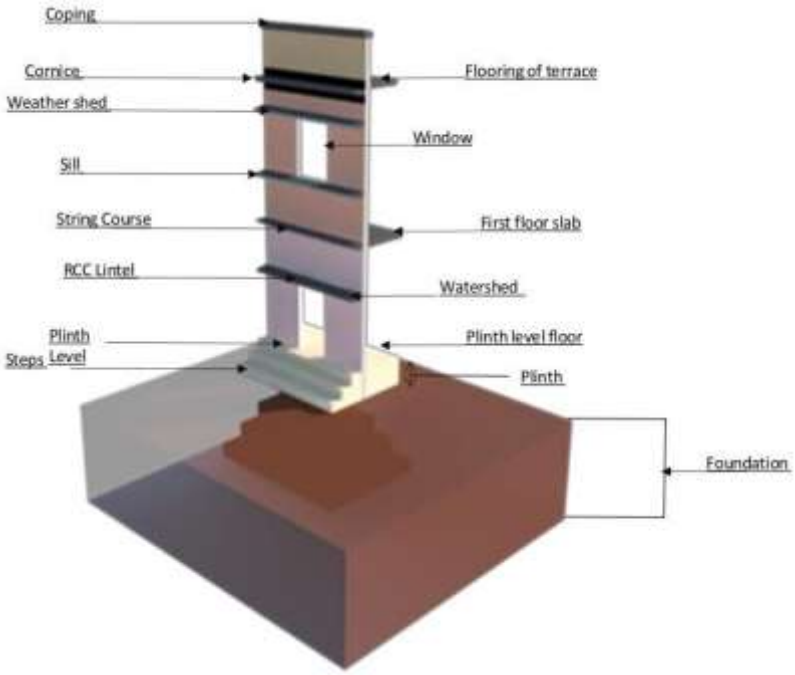
i) Weather Protection: Shields windows and doors from rain, sunlight, and other elements, enhancing indoor comfort.

ii) Sunshade: Reduces heat gain by blocking direct sunlight, improving energy efficiency.

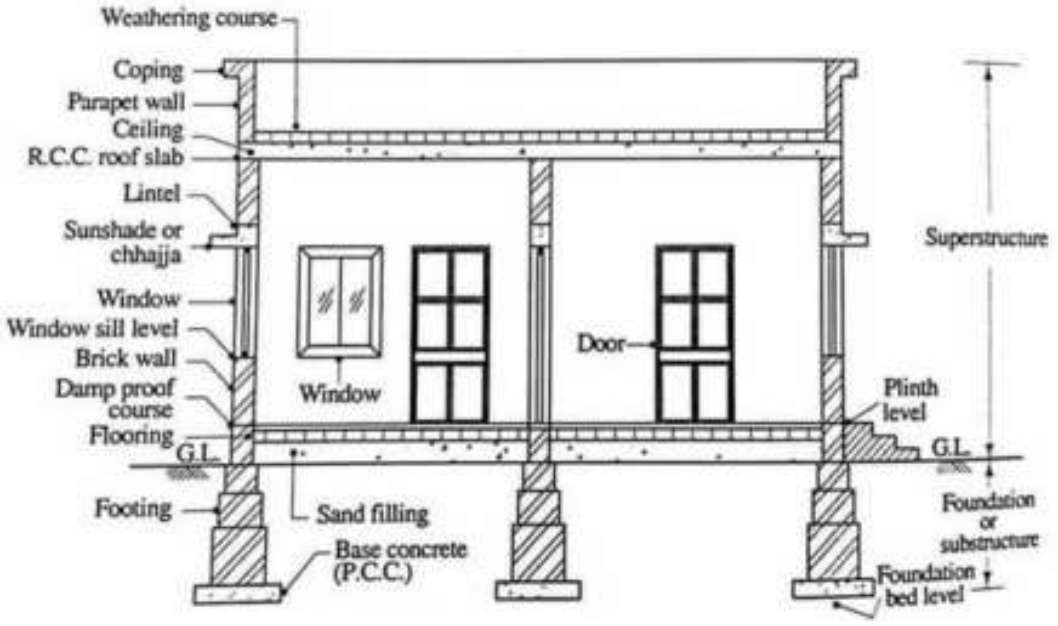
Significance of chejja:

i) Enhances the architectural style of a building while providing practical benefits.

ii) Helps protect the building's facade and prevents water from pooling near the foundation.



Reference: https://civilplanets.com/



Reference: https://housing.com/

Illustrative Representation of different levels on a building

MASONRY WALL

Masonry Wall – Introduction

A wall is a vertical structure, usually made of brick, concrete, or stone, that serves as a physical barrier or divider between spaces. Walls can be found in various settings, including residential, commercial, and industrial buildings, and outdoor environments, such as gardens and parks. In addition to providing a physical barrier, walls can offer security, privacy, and insulation from noise and weather.

Types of Wall Used in Construction:

i) Load Bearing Wall: A load-bearing wall also called a bearing wall supports the roof and upper floors above it. The building would collapse without these walls, which distribute weight to the base. Load-bearing walls usually contain concrete, masonry, or heavy-duty wood framing

ii) Non-Load Bearing Wall: A non-load-bearing inner wall does not support the building above it. It is called a partition wall because it divides a building rather than supporting it. Walls of plasterboard, plaster, or wood framing can be relocated or removed without impacting the building's stability. Non-load-bearing walls can be customized to divide a space into rooms, offices, or storage areas.



Load bearing wall



Non-load bearing wall

Column - Introduction

The most commonly encountered compression member in building constructions is a column. A column is a compression member that transfers load from beam and slab to the structure's foundation. The IS code refers to the column as a compression member, with an effective length 3 times the least lateral dimension.

Types of columns:

There are many distinct kinds of columns that are utilized in various portions of construction. A column is a vertical structural component that primarily supports compression loads. It may distribute the weight from a beam to a floor or foundations, or from a ceiling, floor slab, roof slab, or other slabs. The bending moments about one or both of the cross-section axes are frequently present in columns. The different types of columns based on several factors are listed below.

Based on shape, type of reinforcement, type of loading, on the slenderness ratio and on the type of material

Types of Columns Based on Shape

i) Square/Rectangular columns- These are generally used in building constructions. Due to the ease of shuttering and reinforcement placement, these types of columns are both cost-effective and simple to construct.

ii) Circular columns- Circular columns are commonly used in piling and elevation of buildings. It is also used as bridge pillars. They provide better bending resistance than square or rectangular column

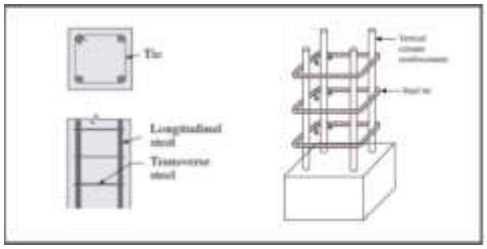
iii) L-type columns- These types of columns are commonly used at the corners of boundary walls.

iv) T-type columns- These types of columns are quite commonly used in bridge construction.

v) Y-type columns- They are used in bridge and flyover construction

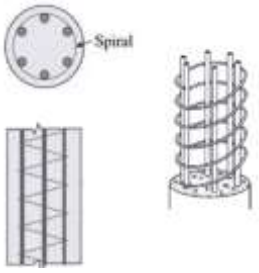
Types of type of reinforcement:

i) Tied Columns- These are the types of columns in which the main longitudinal bars are enclosed within closely and uniformly spaced lateral ties. These are the most commonly used types of reinforced columns



Tied Column

ii) Spiral Columns- In these types of columns, the main longitudinal bars are confined within continuously wound spiral reinforcement. The spiral reinforcements provide lateral support and delay failure due to axial load.



Spiral Column



Composite Column

iii) Composite Columns- These are the types of columns where the reinforcement is in the form of structural steel sections or pipes with or without longitudinal bars.

Types of Columns Based on the Slenderness Ratio

The slenderness ratio of a compression member is defined as the ratio of its effective length to its lateral dimensions. It provides a measure of the column's susceptibility to buckling failure. Columns can be divided into two types of columns based on the slenderness ratio.

i) Short column- The column is referred to as a short column if the ratio of the effective length of the column to the least lateral dimension is less than 12. The failure of a short column is due to crushing (pure compression failure).

ii) Long columns- A long column is defined as one in which the ratio of the effective length of the column to the least lateral dimension is more than 12. Bending or buckling is how a long column fails.

Types of Columns Based on type of loading:

i) Axially Loaded Columns:

- Short Columns: These columns primarily experience axial loads and can carry large loads without significant buckling. They usually fail due to material yielding.
- Long Columns: These are prone to buckling under axial loads. Their design focuses on preventing buckling failure.

ii) Laterally Loaded Columns: These columns are subjected to lateral forces, such as wind or seismic loads. Their design must consider both bending and shear stresses.

iii) Eccentrically Loaded Columns: When loads are applied away from the centroid of the column's cross-section, it creates a bending moment in addition to axial load. This type of loading requires careful design to handle both axial and bending stresses.

Types of Columns based on material:

i) Concrete Columns: Reinforced Concrete Columns: Embedded with steel bars (rebar) to enhance tensile strength, commonly used in buildings and bridges. Precast Concrete Columns: Manufactured off-site and then transported to the construction site, allowing for faster construction.

ii) Steel Columns: Structural Steel Columns: Made from various steel profiles (I-beams, H-sections, etc.), known for high strength-to-weight ratios and ease of fabrication. Hollow Steel Columns: Tubular sections that provide high strength while using less material, often used in high-rise buildings.

iii) Timber Columns: Solid Timber Columns: Made from whole pieces of wood, used in residential buildings and structures with aesthetic considerations. Glulam Columns: Composed of layers of wood glued together, offering greater strength and flexibility than solid wood.

iv) Masonry Columns: Brick Columns: Built from brick or block, often used in low-rise buildings; their strength depends on the quality of materials and construction techniques. Stone Columns: Made from natural stone, providing durability and aesthetic appeal but generally less common in modern construction.

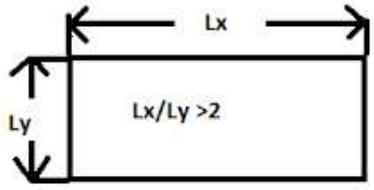
SLABS

Slabs – Introduction

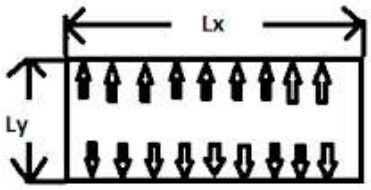
A concrete slab is one of the structural members of buildings or infrastructure. The slab is constructed generally in uniform thickness, but it may vary in some cases. The slab is usually constructed with concrete ingredients. It consists of coarse aggregate, fine aggregate, cement material and structural steel. Steel-reinforced slabs, typically between 100 and 500 mm thick, are most often used to construct floors and ceilings. The concrete slab is constructed supporting the walls, beams and columns of the structures.

Classification of slabs:

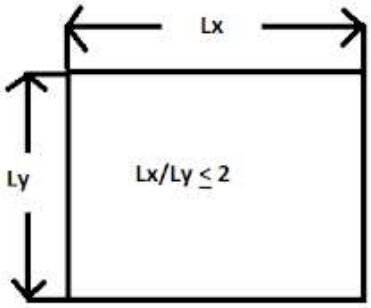
Slabs are generally classified into one-way slab and two-way slab. The former is supported on two sides and the ratio of long to short span is greater than two. However, the latter is supported on four sides and the ratio of long to short span is smaller than two.



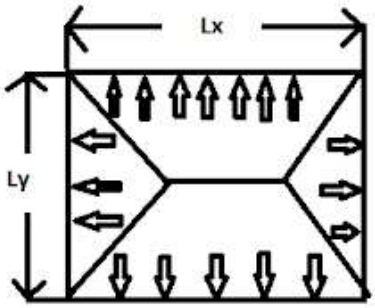
One-way Slab



Load Transfer - One-way Slab



Two-way Slab



Load Transfer - Two-way Slab

Reference: https://www.civillead.com/

Beams - Introduction

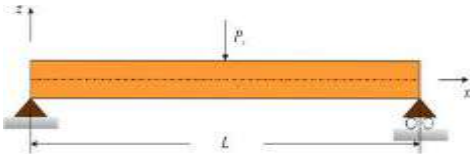
A beam is a structural element or member that largely transfers loads placed along its axis to its supports, such as walls, columns, foundations, and so on, with bending being the primary way of deflections. The longitudinal dimension is predominant relative to cross-section dimension in beam. They are classified based on geometry, support conditions, material, cross section and loading.

i) Based on Geometry

- Straight Beams: The most common type, with a uniform cross-section along their length.
- Curved Beams: Beams that have a curved shape, often used in arches and certain types of bridges.

ii) Based on Support Conditions

- Simply Supported Beams: Supported at both ends, allowing for rotation but not translation.
- Cantilever Beams: Fixed at one end and free at the other, experiencing bending moments due to loads applied to the free end.
- Continuous Beams: Supported on more than two supports, allowing for better load distribution.
- Overhanging Beams: Extend beyond their supports, allowing for loads to be applied at the overhanging section.
- Fixed Beams: Beam having fixed type support on both edges



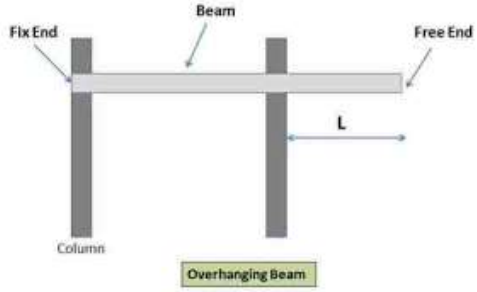
Simply supported beam



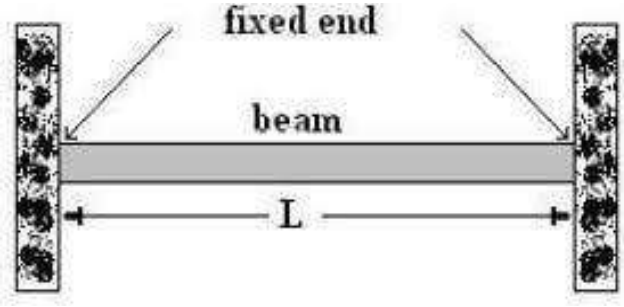
Cantilever beam



Continuous beam



Overhanging beam



Fixed beam

iii) Based on Material

- Steel Beams: High strength and durability; used in various applications, especially in commercial and industrial buildings.
- Reinforced Concrete Beams: Made of concrete and reinforced with steel, commonly used in buildings and bridges for their load-bearing capacity.
- Timber Beams: Made from wood, often used in residential construction for their aesthetic qualities.
- Composite Beams: Combine materials (e.g., steel and concrete) to optimize performance for specific applications.

iv) Based on Cross-Section Shape

- I-Beams: Characterized by their "I" shape, providing high strength with reduced weight.
- T-Beams: Shaped like a "T," commonly used in concrete structures.
- C-Channels: Have a C-shaped cross-section, useful in various structural applications.
- Box Beams: Hollow rectangular or square cross-sections, often used in steel construction.

v) Based on Load Type

- Uniformly Distributed Load (UDL): Loads spread evenly across the beam length.
- Uniformly Varying Load (UVL): Loads varying uniformly across the beam length.
- Point Load: A concentrated load applied at a specific point along the beam

STAIRCASE

Staircase – Introduction

Stairs are the most important components in a building. They are used to access various floors. There are different types of stairs such as spiral stairs, dog legged stairs, open newel stairs, etc, used in commercial, residential buildings, and industrial buildings.

Stairs is a set of steps which give access from floor to floor. The room or enclosure of the building, in which stair is located is known as staircase. Staircase provide access & communication between floors in multi-storey buildings and are a path by which fire can spread from one floor to another. Therefore it must be enclosed by fire resisting walls, floors, ceilings and doors. It must be designed to carry certain loads, which are similar to those used for design of the floors. Stairs may be constructed of Timber, Bricks, Stone, Steel or Reinforced Cement Concrete.

Technical terms related to staircase:

i) Step: It is a portion of stair which permits ascent or descent. A stair is composed of a set of steps.

ii) Tread: It is a upper horizontal portion of a step upon which foot is placed while ascending or descending. Riser: It is a vertical portion of a step providing support to the tread.

iii) Landing: It is level platform at the top or bottom of a flight between the floors.

iv) Flight: This is an unbroken series of steps between landing.

v) Rise: It is a vertical distance between two successive tread faces.

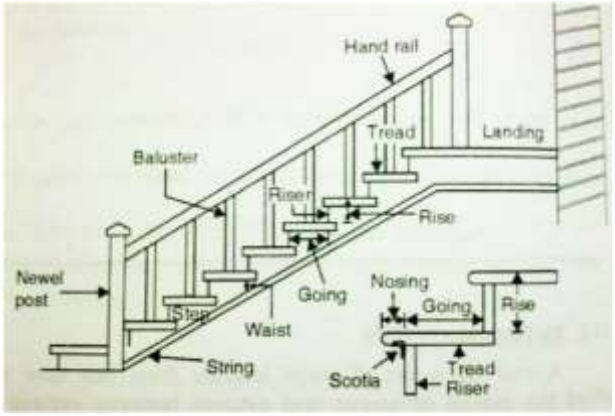
vi) Going: It is a horizontal distance between two successive riser faces.

vii) Nosing: It is the projecting part of the tread beyond the face of riser.

viii) Scotia: It is a moulding provided under the nosing to provide strength to nosing.

ix) Soffit: It is the underside of a stair.

x) Pitch or Slope: It is the angle which the line of nosing of the stair makes with the horizontal.



Components of Staircase

Depending upon the various arrangement of steps, stairs can be classified into:

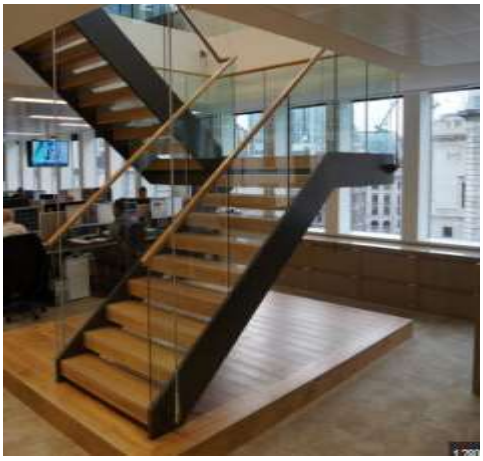
i) Straight Stair: In these types of stairs, all the steps are arranged continuously along in one direction. One flight may be split into one or more than one flight by interposing a landing. This stair can be used where narrow and long space is available for a staircase such as entrance, porch etc.

ii) Dogged-legged Stair: This stair types consist of two straight flights of steps with direct turns between them. Dog-legged stair is very useful where the total width of the stair is just twice the width of the steps.

iii) Open Newel Stair: This type of stairs consists of two or more flights arranging a well or opening between the backward and forward flights. When all the steps are difficult to arrange in two flights, a short third flight of 3 to 6 steps may be provided along the direction perpendicular to the hall. Open newel stair is mostly adopted in the lift.



Straight Stair



Dog-legged Staircase



Open Newel Staircase

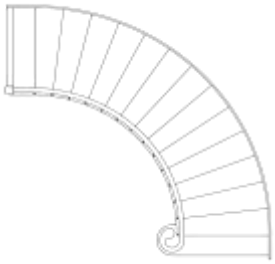
iv) Geometrical Stair: This is another type of open newel stair where the open well between the forward and the backward flight is curved. This stair may contain different geometrical shapes. Here the change in direction is achieved by using winders.

v) Circular Stair: A circular staircase utilizes a large radius coupled with a somewhat smaller radius. These two curves commonly share the same center point. The distance between the two curves (radii) creates the width of the staircase.

vi) Spiral Stair: The spiral stair is very similar to a circular stair. It consists of individual steps or treads, connecting to a center column. The overall diameter of the stair may range from 1 to 2.5 m. Such stairs are provided where space available for stairs is very much limited. Figure shows a typical spiral stair. Cast iron, steel or R.C.C. is used for building these stairs



Geometric staircase



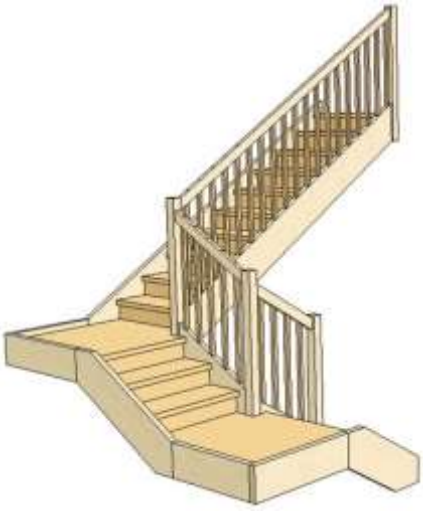
Circular staircase



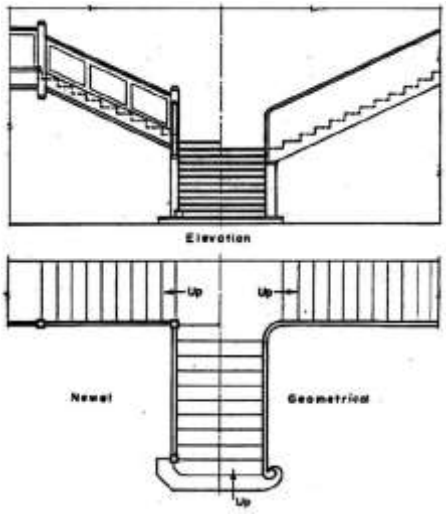
Spiral staircase

vii) Quarter-turn Stair: A quarter turn stair, also known as a 90-degree turn stair, is a type of staircase that makes a 90-degree turn between two flights. This design is often used in buildings where space is limited or to create a more visually aesthetic appealing layout.

viii) Bifurcated Stair: A bifurcated staircase is a type of staircase that features a central flight of stairs that splits into two separate flights at a landing. This design is often used in grand entrances, hotels, and large public buildings to create an elegant and visually appealing focal point.



Quarter turn staircase



Bifurcated staircase

Requirements of Good Staircase:

Location:

i) They should be located near the main entrance to the building.

ii) There should be easy access from all the rooms without disturbing the privacy of the rooms.

iii) There should be spacious approach.

iv) Good light and ventilation should be available.

Width of stair

i) It should be wide enough to carry the user without much crowd on inconvenience.

ii) In Residential building, a 90 cm wide stair is sufficient while in public 1.5 to 1.8 m.

Length of Flight:

i) The number of steps should not be more than 12 & less than 3 from comfort point of view.

Step dimension:

i) The rise and going should be of such dimensions as to provide comfort to users.

ii) The going should not be less than 25 cm, though 30 cm going is quite comfortable.

iii) The rise should be between 10 to 15 cm

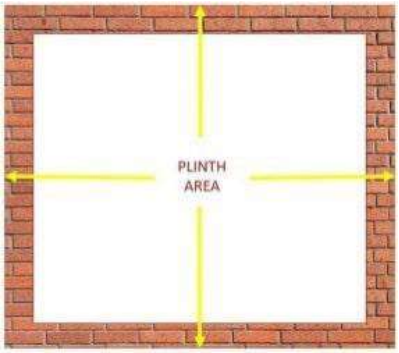
iv) The width of landing should not be less than width of stair.

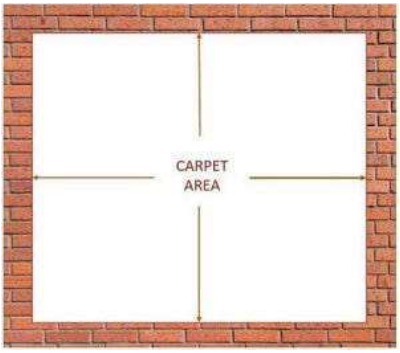
PLINTH AREA & CARPET AREA

Plinth Area & Carpet Area – Introduction

Plinth area = building carpet area + wall area (both internal and exterior walls) + parasitic area + elevator openings, etc. The plinth area is the space between the building's exterior and outer bounds or its walls. The carpet area is the sum of the actual areas of the rooms that can be carpeted.

Plinth area and carpet area of a building is measured for estimation and calculation of building cost. It is also a measure of usable space of building. Plinth area is the covered built-up area measured at the floor level of any storey or at the floor level of the basement. Plinth area is also called as built-up area and is the entire area occupied by the building including internal and external walls. Plinth area is generally 10-20% more than carpet area.

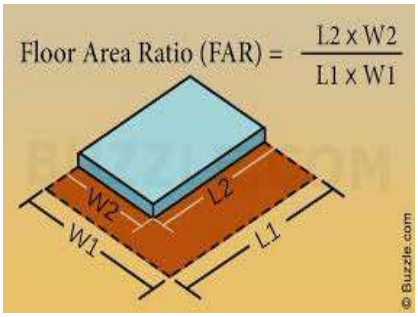




Carpet area the covered area of the usable spaces of rooms at any floor. It is measured between walls to walls within the building and is the sum of the actual areas of the rooms where you can carpet

Floor area ratio – Introduction

Floor Area Ratio (FAR) is a key metric used in urban planning and zoning to determine the relationship between the total building floor area and the size of the lot on which it stands. It is defined as the ratio of the total building floor area (including all floors) to the area of the lot or parcel of land. It helps in regulating building density in a specific area.



$$Floor\ Area\ Ratio\ (FAR) = \frac{Total\ Floor\ Area\ of\ the\ Building}{Total\ Plot\ Area}$$

LOCAL BUILDING BYELAWS

Local building byelaws – Introduction

Building bye-laws, also known as building codes or regulations, are a set of rules and guidelines established by local, regional, or national authorities to govern the construction, renovation, and maintenance of buildings. These bye-laws are essential for ensuring the safety, structural integrity, and functionality of structures and promoting the well-being of occupants and the environment. However, the specifics of building bye-laws can vary significantly from one jurisdiction to another, reflecting local conditions, cultural preferences, and technological advancements.

The recent trend in the building byelaws emphasis on the concept of sustainability focus on green building practices, energy efficiency, sustainable materials and digitization of approval process with adoption of online platforms for plan submissions and approvals to streamline the process.

Purpose of Byelaws:

i) Safety: Ensure structural integrity and safety from natural hazards.

ii) Health and Hygiene: Promote adequate ventilation, sanitation, and waste management.

iii) Aesthetics and Urban Planning: Regulate building heights, architectural styles, and land use to maintain the character of neighborhoods.

iv) Environmental Protection: Encourage sustainable practices and manage environmental impact

Key components of Byelaws:

i) Site Development: Guidelines for site coverage, setbacks, open spaces, and landscaping.

ii) Building Height and FSI: Regulations on the maximum height of buildings and Floor Space Index (FSI), determining the permissible built-up area.

iii) Use of Buildings: Zoning regulations that define permissible land use (residential, commercial, industrial, etc.).

iv) Building Design and Construction: Standards for materials, structural design, safety features (like fire safety), and accessibility.

v) Building Services: Requirements for plumbing, electrical systems, waste disposal, and ventilation.

vi) Parking and Traffic Management: Specifications for parking space requirements based on building use and size.

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